

Nonintegral Laplacian spectra of ideal intersection graphs

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Abstract

We study Laplacian integrality for the graph of nonzero proper ideals of $\mathbb{Z}_n$, where distinct ideals are adjacent when their intersection is nonzero. For squarefree composite n , the graph is Laplacian integral exactly when n has two prime factors. For three distinct primes we prove nonintegrality for every exponent vector (a, a, b) with $a, b \geq 1$, including $(1, 1, k)$, and for every three-prime exponent vector with minimum entry at most three. For arbitrary ordered exponents $2 \leq a \leq b \leq c$, we also prove nonintegrality whenever $c \geq 4a^2 - 2a$. The repeated-exponent proof combines arithmetic constraints on a quadratic discriminant with a uniform consecutive-integer interval for a symmetric cubic. Exact divisor reductions settle the few exceptional factor levels. Complement-quotient intervals prove the families with an exponent equal to one, two or three; the last case uses an exhaustive integer endpoint certificate after the uniform cutoff. Three pairwise distinct exponents $4 \leq a < b < c < 4a^2 - 2a$, and nonsquarefree integers with more prime factors remain open.

1 Introduction

The ideal intersection graph associates a graph to the intersection structure of the nonzero proper ideals of a ring [2]. For $\mathbb{Z}_n$, its isomorphism type depends on the exponents in the prime factorization of n . We ask which of these graphs have integer Laplacian eigenvalues.

Nikandish [1] develops a generalized-join formula for the Laplacian spectrum and a Schur–Weyl reduction for equal prime exponents. His results include nonintegrality for $(pqr)^k$ for every $k \geq 1$. His Question (Q3) asks for the full integrality characterization; the present arguments address this question for additional exponent families.

We prove a complete classification for squarefree composite integers (Theorem 3.1), an explicit quadratic obstruction for $(1, 1, k)$ (Theorem 4.1), and nonintegrality for every repeated-exponent triple (a, a, b) , $a, b \geq 1$ (Theorem 7.1). The last result combines an arithmetic restriction on the antisymmetric block with a consecutive-integer sign interval for the symmetric cubic. The boundary family $b = (a - 1)(2a - 1)$ is proved in Theorem 5.1. An exact divisor criterion isolates every possible square discriminant for fixed a . A second divisor criterion fixes the smaller-factor index instead, giving a finite reduction over all a (Proposition 6.6). The first three nonboundary factor levels have complete finite modular certificates. Every later level lies strictly below the uniform sign threshold, closing the whole repeated-exponent family.

For a unit exponent, the complement of the general six-support quotient has a nonzero eigenvalue strictly between 0 and 1, except for three small pairs already covered by $(1, 1, k)$. This proves nonintegrality for every $(1, b, c)$ (Theorem 8.1), including fully unequal triples. Positive-coefficient expansions at a minimum exponent of two, together with one exceptional rational sign interval, then settle every $(2, b, c)$ (Theorem 9.1). Thus every three-prime exponent vector with minimum entry at most two is covered. The same endpoint polynomial gives a uniform cutoff $c \geq 4a^2 - 2a$ for

ordered $2 \leq a \leq b \leq c$ (Proposition 10.1). For a minimum exponent of three, this leaves exactly 325 pairs. An exhaustive integer sign certificate, positive second-endpoint expansions and one rational exception settle all of them (Theorem 10.2). Thus every triple with minimum entry at most three is nonintegral.

The graph and lifting arguments are given explicitly, so the proofs do not require a complete spectral decomposition. Section 12 describes the exact computations supporting the displayed formulas and finite certificates. The classification for arbitrary exponent vectors remains open.

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^t p_i^{k_i}$, where the primes are distinct and $k_i \geq 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \leq \alpha_i \leq k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \neq \emptyset$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H ; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices). *If $L(H)v = \lambda v$ and $\lambda \neq 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \neq 0$ implies $\sum v = 0$. At a vertex of H , its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. $\square$

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2 + ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a + 2ab & -2ab \\ -a & -b & -a^2 & a^2 + a + b \end{pmatrix}. \quad (1)$$

For example, an X vertex has a^2 neighbors in Z and ab in W , while a W vertex has a, b, a^2 neighbors in X, Y, Z . Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2 + ab & -ab \\ -a & a^2 + a + b + 2ab \end{pmatrix}. \quad (2)$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = xf_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a + b + 1)(a^2 + 2ab + 2a + b). \end{aligned} \quad (3)$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n . A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Squarefree integers

Theorem 3.1. *For squarefree composite n , the graph G_n is Laplacian integral if and only if n has exactly two prime factors.*

Proof. Let t be the number of prime factors. For $t = 2$ the graph consists of the two ideals (p) and (q) , with no edge, so $L = 0$. Assume $t \geq 3$. The vertices are the nonempty proper subsets of $[t]$, adjacent when they intersect. On the tensor space encoding all subsets put

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = 2^t - 1.$$

The principal submatrix of $\Phi = cI - E^{\otimes t} + F^{\otimes t} - J^{\otimes t}$ on the graph vertices is $L(G_n)$: the diagonal entry at S is $2^t - 2 - 2^{t-|S|}$, and the off-diagonal entry is -1 for intersecting subsets and zero otherwise.

With e_0, e_1 denoting the absence and presence basis, set $w = e_0 \otimes e_1 - e_1 \otimes e_0$. For every 2×2 matrix T , one has $T^{\otimes 2}w = (\det T)w$. A paired factor w thus contributes $2, -1, 0$ under E, F, J , respectively. It vanishes at the empty and full subsets. Each subspace used below is invariant for Φ and is supported on graph vertices, so its restriction is also a restriction of L .

If $t = 2r + 1$, $r \geq 1$, use $w^{\otimes r} \otimes \text{span}\{e_0, e_1\}$. Putting $a = 2^r$ and $s = (-1)^r$, the restricted matrix is

$$\begin{pmatrix} c - 2a + s & s \\ s & c - a \end{pmatrix},$$

with discriminant $(a - s)^2 + 4$. The integer $h = a - s$ is at least 3, so $h^2 < h^2 + 4 < (h + 1)^2$. Both eigenvalues are irrational.

If $t = 2r$, $r \geq 2$, use $r - 1$ paired factors and the symmetric subspace on the last two coordinates, with basis $e_0e_0, (e_0e_1 + e_1e_0)/\sqrt{2}, e_1e_1$. Put $a = 2^{r-1}$ and $s = (-1)^{r-1}$. The restriction is $cI - K$, where

$$K = \begin{pmatrix} 4a - s & -s\sqrt{2} & -s \\ -s\sqrt{2} & 2a - s & 0 \\ -s & 0 & a \end{pmatrix}.$$

The leading principal minors of $K - (a - 1)I$ are

$$3a + 1 - s, \quad (3a + 1 - s)(a + 1 - s) - 2, \quad (3a - s)(a + 1 - s) - 2.$$

They are positive for $a \geq 2$ and $s = \pm 1$: the latter two are at least 10 and 8, respectively. Sylvester's criterion gives $\lambda_{\min}(K) > a - 1$. The principal submatrix of $K - aI$ on coordinates 1, 3 is $\begin{pmatrix} 3a-s & -s \\ -s & 0 \end{pmatrix}$, with determinant -1 . A vector supported on those coordinates has a negative Rayleigh quotient, giving $\lambda_{\min}(K) < a$. Hence $c - \lambda_{\min}(K)$ is an eigenvalue of L strictly between the consecutive integers $c - a$ and $c - a + 1$. $\square$

4 The family pqr^k

Theorem 4.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k + 1 \pm \sqrt{(2k + 2)^2 - 3}}{2}.$$

In particular, every permutation of the exponent vector $(1, 1, k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes 1, 2 have size 1, and 13, 23 have size k . Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k + 1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k + 1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k + 1)x + 8k^2 + k$, with discriminant $(2k + 2)^2 - 3$. Since

$$(2k + 1)^2 < (2k + 2)^2 - 3 < (2k + 2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. $\square$

5 A boundary family with integral antisymmetric block

The following argument was supplied by Reza Nikandish in the public collaboration discussion. We include it because it is the boundary case needed for the linear cutoff in Section 6.

Theorem 5.1. *For every $a \geq 2$, put $C = (a - 1)(2a - 1)$. For distinct primes p, q, r , the graph $G_{p^a q^a r^C}$ is not Laplacian integral, although the antisymmetric block (2) has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at 0, 1, 2, 3, 4 modulo 5 are 3, 4, 2, 3, 3. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$\begin{aligned} f_{a,C}(m) &= 2(5a^4 - 10a^3 + 7a^2 - 1), \\ f_{a,C}(m - 1) &= -2(a - 1)(a + 2)(2a^4 - 7a^3 + 7a^2 - a - 3). \end{aligned}$$

The first value is positive because its parenthesis is $5a^3(a-2)+7a^2-1$. Writing $z = a-3 \geq 0$, the last factor in the second value is $2z^4+17z^3+52z^2+68z+30 > 0$. Consequently $f_{a,C}(m-1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m-1$ and m . Its integer shift by $u = a^2C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a+1)^2b^2 + 2a(3a+1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. $\square$

6 Finite reduction and a linear cutoff

Fix $a, b \geq 1$, and put

$$h = a + 1, \quad C = (a-1)(2a-1), \quad M = a^3(2a+1), \quad A = h^2b + a(3a+1).$$

The characteristic discriminant of (2) is

$$D = h^2b^2 + 2a(3a+1)b + a^2. \quad (4)$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2b - 1$.

Proposition 6.1 (Exact divisor reduction). *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a+1)}{h^2} \in \mathbb{Z}_{\geq 1}, \quad s = \frac{e - d}{h} \in \mathbb{Z}_{\geq 0}. \quad (5)$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2D = 4M \quad (6)$$

holds for arbitrary a, b . If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (5). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d + e)^2 - (e - d)^2 = 4de = 4M$ in (6) gives $D = s^2$. $\square$

The divisor set is finite for every fixed a ; this criterion has no truncation in b . It gives first a quadratic bound.

Corollary 6.2. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d-1)(e-1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a+1) = h^2C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. $\square$

Lemma 6.3. *Every admissible factor pair in Proposition 6.1 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a+1) \equiv 2 \pmod{h}$. Also

$$(d-1)(e-1) = M - A + 1 = h^2(C-b). \quad (7)$$

If h is odd, the congruence $2(d-1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d-1 \equiv h/2 \pmod{h}$. Then $d-1$ and $e-1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (7). This excludes the alternative for either parity of a . $\square$

Theorem 6.4 (Unified linear cutoff). *Let $a \geq 2$ and*

$$R(a) = \frac{2(a-1)(a-2)}{a+2} = 2a - 10 + \frac{24}{a+2}.$$

For every integer $b > R(a)$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved repeated-exponent pair therefore has square D and $1 \leq b \leq \lfloor R(a) \rfloor$.

Proof. If D is nonsquare the antisymmetric block suffices. Otherwise use Proposition 6.1. When $d = 1$, $e = M$ and $b = C$; Theorem 5.1 settles this case. If $d > 1$, Lemma 6.3 gives $d \geq T = a + 2$. Since $e \geq d \geq T$,

$$\frac{M}{T} + T - (d+e) = (d-T) \left(\frac{e}{T} - 1 \right) \geq 0.$$

Hence

$$b \leq \frac{M/T + T - a(3a+1)}{h^2} = \frac{2(a-1)(a-2)}{a+2} = R(a).$$

Thus $b > R(a)$ leaves either a nonsquare discriminant or the already settled boundary case. $\square$

At $(a, b) = (4, 2)$ the square-discriminant pair has $d = 6 = a + 2$, so equality in the factor estimate is attained. The graph is still nonintegral, as shown in the next section.

6.1 Settling the first nonboundary factor

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M , the identity

$$M = (a+2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a+2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (5) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive- b list and its cubic certificates are:

(a, b)	$f_{a,b}(x)$	Root-free prime
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(6, 5)$	$x^3 - 245x^2 + 19266x - 488160$	7
$(10, 12)$	$x^3 - 842x^2 + 230160x - 20534400$	13
$(22, 35)$	$x^3 - 4919x^2 + 7887858x - 4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d = a + 2$ pair is settled, over all a , by this finite divisor argument.

Theorem 6.5 (Second linear cutoff). *For $a \geq 2$ put*

$$S(a) = \frac{(a-3)(2a-3)}{2a+3} = a - 6 + \frac{27}{2a+3}.$$

For every positive integer $b > S(a)$ and distinct primes p, q, r , $G_{p^a q^a r^b}$ is not Laplacian integral. Every remaining square-discriminant case has $d \geq 2a+3$ and $b \leq \lfloor S(a) \rfloor$.

Proof. Nonsquare D is covered by the antisymmetric block. If D is square, the cases $d = 1$ and $d = a + 2$ are already settled. Every other admissible factor satisfies $d \geq T = 2a + 3$ by Lemma 6.3. Since $e \geq d \geq T$, the same factor inequality gives

$$b \leq \frac{M/T + T - a(3a+1)}{(a+1)^2} = \frac{(a-3)(2a-3)}{2a+3} = S(a).$$

This proves the claim for $b > S(a)$. □

The new bound is strictly smaller than $R(a)$ for all $a \geq 2$, since

$$R(a) - S(a) = \frac{2a^3 - a^2 - a - 6}{(a+2)(2a+3)} > 0.$$

Writing $z = a - 2 \geq 0$, its numerator is $2z^3 + 11z^2 + 19z + 4$.

6.2 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d = 1 + hj$, $j \geq 1$. The following criterion fixes j instead of a .

Proposition 6.6. *Fix $j \geq 1$ and set $K_j = (j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a = (d-1)/j - 1 \in \mathbb{Z}_{\geq 2}$. Set $e = M/d$, $v = (e-1)/(a+1)$, and retain exactly those candidates with $d \leq e$ and $b = C - jv \geq 1$. Then e, v, b are integers and $D = (v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d = j(a+1) + 1$ we have $ja \equiv -(j+1)$, whence

$$j^4 M = (ja)^3 j(2a+1) \equiv (j+1)^3(j+2) = K_j \pmod{d}.$$

Since $\gcd(d, j) = 1$, we have $d \mid M$ if and only if $d \mid K_j$. Thus e is integral. Since $M \equiv d \equiv 1 \pmod{h}$, also $e \equiv 1 \pmod{h}$, so v is integral. The identity

$$M = 1 + h(3a-2) + h^2 C = (1+hj)(1+hv)$$

gives $j + v + hjv = 3a - 2 + hC$. Defining $b = C - jv$, we obtain $d + e = h^2 b + a(3a+1)$ and $e - d = h(v-j)$. The ordering condition gives $v-j \geq 0$; Proposition 6.1 now proves admissibility and $D = (v-j)^2$. The reverse direction follows by applying these identities to any admissible index- j pair. □

This is a complete finite problem for each fixed j , without a truncation in a . Section 7 combines the first three such problems with a uniform argument for all later indices. For $j = 2$, $K_2 = 108$ and $d = 2a + 3$ is an odd divisor of 108 with $d \geq 7$. The only candidates are $d = 9, 27$, giving $a = 3, 12$. The first has $e = 21, v = 5, b = 0$ and is excluded. The second has $e = 1600, v = 123, b = 7$ and $\sqrt{D} = 121$. Thus $(12, 7)$ is the only positive- b case with $j = 2$.

Its symmetric cubic is $f_{12,7}(x) = x^3 - 751x^2 + 180348x - 13829760$. Its values at all residues modulo 13 are 4, 7, 9, 3, 8, 4, 10, 6, 11, 5, 7, 10, 7. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

Theorem 6.7 (Third linear cutoff). *For every $a \geq 2$, put $T(a) = 2(a-4)(a-2)/(3a+4)$. If $b \geq 1$ and $b > T(a)$, then $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved square-discriminant pair has $j \geq 3$, $d \geq 3a+4$, and $b \leq \lfloor T(a) \rfloor$.*

Proof. Nonsquare D gives irrational antisymmetric eigenvalues. For square D , the cases $j = 0, 1, 2$ are all settled above. Every other pair has $d \geq Q = 3a+4$, and $e \geq d \geq Q$ gives

$$b \leq \frac{M/Q + Q - a(3a+1)}{(a+1)^2} = \frac{2(a-4)(a-2)}{3a+4} = T(a).$$

This proves the claimed implication. □

For $a \geq 3$ the new bound is strictly below $S(a)$, since

$$S(a) - T(a) = \frac{2a^3 - a^2 - 5a - 12}{(2a+3)(3a+4)} > 0.$$

With $z = a - 3 \geq 0$, the numerator is $2z^3 + 17z^2 + 43z + 18$. The exact expression $T(a) = 2a/3 - 44/9 + 320/(9(3a+4))$ shows its leading term.

7 Completion of the repeated-exponent family

Theorem 7.1. *For distinct primes p, q, r and all integers $a, b \geq 1$, the graph $G_{p^a q^a r^b}$ is not Laplacian integral. The same conclusion holds for every permutation of the exponent vector (a, a, b) .*

The arithmetic reduction and a uniform sign obstruction for the cubic combine to exhaust the family. We first settle one last finite factor level.

Lemma 7.2. *Every square-discriminant pair with index $j = 3$ gives a nonintegral Laplacian spectrum.*

Proof. Proposition 6.6 gives $d \mid K_3 = 320$ and $d = 3a+4$. For $a \geq 2$, the complete divisor list of this shape is 10, 16, 40, 64, 160. The resulting values are

d	a	e	v	b	Outcome
10	2	4	1	0	$d > e$
16	4	36	7	0	$b = 0$
40	12	1080	83	4	Root-free modulo 5
64	20	5125	244	9	Root-free modulo 7
160	52	92274	1741	30	Root-free modulo 7

The three positive- b cubics are

$$\begin{aligned} f_{12,4}(x) &= x^3 - 568x^2 + 100032x - 5248512, \\ f_{20,9}(x) &= x^3 - 1769x^2 + 992820x - 174744000, \\ f_{52,30}(x) &= x^3 - 13394x^2 + 57771168x - 80229951360. \end{aligned}$$

Their values at all residues of the indicated primes are, respectively, $(3, 3, 3, 4, 2)$, $(3, 2, 4, 1, 6, 4, 1)$ and $(1, 6, 4, 1, 3, 2, 4)$. Each monic cubic is therefore irreducible over $\mathbb{Q}$ and has noninteger real roots. Section 2 supplies the lift to noninteger graph eigenvalues. □

Lemma 7.3 (A uniform consecutive-integer interval). *For $a, b \geq 1$, put*

$$L(a) = \frac{a^3 + 2a^2 + 2a + 1}{a(2a + 1)}.$$

If $b < L(a)$, then $f_{a,b}$ has a root strictly between $2ab - 1$ and $2ab$.

Proof. Substitution in (3) gives

$$\begin{aligned} f_{a,b}(2ab) &= 2a^2b^2 > 0, \\ f_{a,b}(2ab - 1) &= -(a + b + 1)(a^3 + 2a^2 + 2a + 1 - a(2a + 1)b). \end{aligned}$$

The second value is strictly negative when $b < L(a)$. The intermediate value theorem gives the asserted noninteger, nonzero root. $\square$

Proof of Theorem 7.1. For $a = 1$, Corollary 6.2 settles every positive b . For $a \geq 2$, nonsquare D gives irrational antisymmetric eigenvalues. When D is square, index $j = 0$ is Theorem 5.1, and indices $j = 1, 2$ are settled in Section 6. Lemma 7.2 settles $j = 3$.

It remains to consider $j \geq 4$. Then $d \geq Q = 4a + 5$, and $e \geq d \geq Q$ implies

$$\frac{M}{Q} + Q - (d + e) = (d - Q) \left(\frac{e}{Q} - 1 \right) \geq 0.$$

Since $d + e = (a + 1)^2b + a(3a + 1)$, we obtain

$$b \leq B_4(a) = \frac{M/Q + Q - a(3a + 1)}{(a + 1)^2} = \frac{(a - 5)(2a - 5)}{4a + 5}.$$

For every $a \geq 2$,

$$L(a) - B_4(a) = \frac{41a^3 - 17a^2 - 11a + 5}{a(2a + 1)(4a + 5)} > 0.$$

Indeed, with $z = a - 2 \geq 0$, the numerator becomes $41z^3 + 229z^2 + 413z + 243$. Thus $b \leq B_4(a) < L(a)$, and Lemma 7.3 gives a noninteger cubic root, which lifts across the universal vertices. These cases exhaust all positive a, b . Permuting the prime factors preserves the graph's isomorphism type. $\square$

The preceding cutoffs and fixed-exponent theorems are intermediate consequences of the arithmetic reduction. The uniform interval now settles the infinite range left by those cutoffs, without a parameter scan.

8 A unit exponent with two arbitrary exponents

Theorem 8.1. *For distinct primes p, q, r and every $b, c \geq 1$, the graph $G_{pq^br^c}$ is not Laplacian integral. The same holds for permutations of the exponent vector $(1, b, c)$.*

The equal-exponent decomposition is replaced by a complement identity on all six support classes. For arbitrary $a, b, c \geq 1$, remove the $u = abc - 1$ universal vertices and denote the remaining graph by H . In the order $1, 2, 3, 12, 13, 23$, its class-size vector is $w = (a, b, c, ab, ac, bc)^T$ and its

vertex count is $N = a + b + c + ab + ac + bc$. Disjoint supports are adjacent in the complement $\overline{H}$, whose cell-constant Laplacian quotient is

$$C = \begin{pmatrix} b+c+bc & -b & -c & 0 & 0 & -bc \\ -a & a+c+ac & -c & 0 & -ac & 0 \\ -a & -b & a+b+ab & -ab & 0 & 0 \\ 0 & 0 & -c & c & 0 & 0 \\ 0 & -b & 0 & 0 & b & 0 \\ -a & 0 & 0 & 0 & 0 & a \end{pmatrix}.$$

Multiplication on the left by $\text{diag}(w)$ makes C symmetric. Moreover $C\mathbf{1} = 0$ and $w^T C = 0$. If B is the quotient of H , the complement relation gives

$$B + C = NI - \mathbf{1}w^T. \quad (8)$$

Thus an eigenvector $Cv = \mu v$ with $\mu \neq 0$ satisfies $w^T v = 0$ and $Bv = (N - \mu)v$. When $N - \mu \neq 0$, Lemma 2.1 lifts it to an eigenvalue $N + u - \mu$ of G_n .

Proof of Theorem 8.1. Write $\det(xI - C) = xh_{a,b,c}(x)$. Direct determinant expansion yields

$$\begin{aligned} h_{a,b,c}(0) &= -abc(a + b + c)N, \\ h_{a,b,c}(a) &= -a^2b^2c^2(a^2 + 2a - b - c). \end{aligned}$$

At $a = 1$, the first value is strictly negative and

$$h_{1,b,c}(1) = b^2c^2(b + c - 3).$$

When $b + c > 3$, a real root μ lies strictly between 0 and 1. Equation (8) and the universal-vertex lift produce a Laplacian eigenvalue $V - \mu \in (V - 1, V)$, where $V = N + u$ is the integer vertex count. This eigenvalue is noninteger. When $b + c \leq 3$, the positive pairs are $(1, 1), (1, 2), (2, 1)$; Theorem 4.1 settles the corresponding permutations of $(1, 1, k)$. Permuting the prime factors preserves the graph's isomorphism type. $\square$

The general six-support reduction also gives an exact diagnostic for fully distinct triples. For $1 \leq a < b < c \leq 7$, all thirty-five quintics have a nonlinear irreducible factor over $\mathbb{Q}$. The quintics at $(1, 2, 7)$ and $(2, 3, 5)$ have factor degrees $(1, 4)$, and that at $(2, 4, 6)$ has factor degrees $(2, 3)$; irreducibility of the entire quintic is therefore not a uniform obstruction. Theorem 8.1 covers every triple beginning with 1 without a range restriction; the thirty-five-case calculation itself remains finite evidence.

9 A minimum exponent of two

Theorem 9.1. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^2q^brc}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most two gives a nonintegral graph.*

Proof. Use the complement quotient and quintic from Section 8. For $a = 2$, $h_{2,b,c}(0) < 0$. Putting $s = b + c$ and $t = bc$ gives

$$h_{2,b,c}(1) = (s - 13)t^2 - (2s + 6)t + 3s^3 + s^2 - 3s - 1.$$

By symmetry order $b \leq c$. A unit exponent is covered by Theorem 8.1; if $b = 2$ or $b = c$, use Theorem 7.1. It remains to consider $3 \leq b < c$.

When $b = 3$ and $c = 5 + v$, $v \geq 0$, the exact expansion is

$$h_{2,3,5+v}(1) = 12v^3 + 112v^2 + 268v + 120 > 0.$$

When $b \geq 4$ and $c > b$, write $b = 4 + u$ and $c = 5 + u + v$, with $u, v \geq 0$. Then

$$\begin{aligned} h_{2,4+u,5+u+v}(1) &= (u^2 + 8u + 19)v^3 \\ &\quad + (4u^3 + 38u^2 + 128u + 170)v^2 \\ &\quad + (5u^4 + 62u^3 + 271u^2 + 502u + 368)v \\ &\quad + 2u^5 + 32u^4 + 190u^3 + 504u^2 + 552u + 160 > 0. \end{aligned}$$

All coefficients are positive. In either region the opposite signs at 0 and 1 give a root $\mu \in (0, 1)$ and, by (8) and the universal-vertex lift, a noninteger graph eigenvalue in $(V - 1, V)$.

The only remaining pair is $(b, c) = (3, 4)$. Here

$$h_{2,3,4}(x) = x^5 - 53x^4 + 953x^3 - 6523x^2 + 13134x - 7560,$$

and $h_{2,3,4}(1) = -48$ while $h_{2,3,4}(3/2) = 13437/32 > 0$. A root lies in $(1, 3/2)$. Since $V = 58$, its lifted graph eigenvalue lies in $(113/2, 57)$, an interval containing no integer. This exhausts all positive b, c . Combining with Theorem 8.1 proves the minimum-entry assertion. $\square$

10 A uniform cutoff and a minimum exponent of three

Proposition 10.1. *For distinct primes p, q, r and positive exponents ordered as $2 \leq a \leq b \leq c$, the graph $G_{p^a q^b r^c}$ is not Laplacian integral whenever $c \geq 4a^2 - 2a$.*

Proof. Use the complement quintic $h = h_{a,b,c}$ of Section 8. Its first endpoint is the cubic

$$h(1) = Lc^3 + Qc^2 + Rc + S, \tag{9}$$

where

$$\begin{aligned} L &= a^2 + b^2 - 1, \\ Q &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ R &= (a - 1)(b - 1)(2ab + 3a + 3b - 3), \\ S &= (a - 1)(b - 1)(a + b - 1)(ab + a + b - 1). \end{aligned}$$

Here $L, R, S > 0$. In fact the sharper condition $Lc + Q \geq 0$ already gives $h(1) > 0$. Put $T(a) = 4a^2 - 2a$. Then

$$Q + T(a)L = b^2(b - 1) + (2a^2 + a - 3)b + K(a),$$

where $K(a) = 4a^4 - a^3 - 5a^2 - a + 3$. For $a = 2 + u$, $u \geq 0$,

$$K(2 + u) = 4u^4 + 31u^3 + 85u^2 + 95u + 37 > 0.$$

Consequently $Lc + Q > 0$ for $c \geq T(a)$. Since $h(0) < 0$, a root lies in $(0, 1)$, producing a noninteger graph eigenvalue in $(V - 1, V)$ by (8) and Lemma 2.1. $\square$

Theorem 10.2. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^3 q^b r^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most three gives a nonintegral graph.*

Proof. Theorems 8.1, 9.1 and 7.1 leave $4 \leq b < c$. Proposition 10.1 settles $c \geq 30$. At $a = 3$, equation (9) is

$$h_{3,b,c}(1) = (b^2 + 8)c^3 + (b - 1)(b^2 - 30b - 12)c^2 + 6(b - 1)(3b + 2)c + 4(b - 1)(b + 2)(2b + 1).$$

The remaining domain $4 \leq b < c < 30$ has exactly $\binom{26}{2} = 325$ pairs. Exhaustive integer evaluation of this cubic gives 257 positive and 68 negative values, with no zero. Table 1 lists every negative range; all other pairs in this finite domain have positive values. The complete polynomial and separate Horner sign certificates are in `results/distinct-tail.json`; their construction is described in Section 12. Thus a positive value gives a root in $(0, 1)$.

b	4	5	6	7	8	9	10	11	12	13
$c_{\min}$	5	6	7	8	9	10	11	12	13	14
$c_{\max}$	13	15	16	17	17	16	16	15	14	14

Table 1: Every negative $h_{3,b,c}(1)$ in $4 \leq b < c < 30$; each column includes all integers between its two endpoints.

For the negative values, the second endpoint is positive except at $(b, c) = (4, 5)$. Indeed, for $v \geq 0$,

$$h_{3,4,6+v}(2) = 104v^3 + 1238v^2 + 4018v + 2344 > 0.$$

For $b \geq 5$, write $b = 5 + u$, $c = 6 + u + v$, with $u, v \geq 0$. Direct expansion gives

$$\begin{aligned} h_{3,5+u,6+u+v}(2) &= (5u^2 + 54u + 153)v^3 \\ &\quad + (20u^3 + 262u^2 + 1179u + 1863)v^2 \\ &\quad + (25u^4 + 426u^3 + 2642u^2 + 7011u + 6624)v \\ &\quad + 10u^5 + 218u^4 + 1828u^3 + 7200u^2 + 12600u + 6480 > 0. \end{aligned}$$

These regions exhaust $4 \leq b < c$ except $(4, 5)$. Whenever $h(1) < 0 < h(2)$, a root lies in $(1, 2)$, giving a noninteger graph eigenvalue in $(V - 2, V - 1)$.

Finally

$$h_{3,4,5}(x) = x^5 - 83x^4 + 2327x^3 - 24133x^2 + 60576x - 42480.$$

Here $h(1) = -3792$ and $h(3/2) = 48825/32 > 0$. The root in $(1, 3/2)$ lifts to a graph eigenvalue in $(233/2, 117)$, since $V = 118$. Every interval used contains no integer. The finite nonvanishing certificate and the written infinite-range arguments exhaust all b, c ; Theorems 8.1 and 9.1 give the final assertion. $\square$

11 Complete fixed-exponent families and exact diagnostics

Theorem 11.1. *For every $a \in \{2, 3, 4\}$, every $b \geq 1$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral.*

Proof. The cutoff values are $R(2) = 0$, $R(3) = 4/5$, and $R(4) = 2$. Theorem 6.4 settles all b for $a = 2, 3$, and all $b > 2$ for $a = 4$. At $(a, b) = (4, 1)$, $D = 145$ is nonsquare. At $(4, 2)$ the symmetric cubic is

$$f_{4,2}(x) = x^3 - 86x^2 + 2304x - 18816.$$

Its values at $0, 1, 2, 3, 4$ modulo 5 are $4, 3, 1, 4, 3$, so it has no root modulo 5. A cubic over a field without a root is irreducible. Since this cubic is monic, reduction modulo 5 proves irreducibility over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues. $\square$

Corollary 11.2. *The same all- b conclusion holds for $a = 5, 6$. Thus it holds for every $a \in \{2, 3, 4, 5, 6\}$.*

Proof. The second cutoff gives $S(5) = 14/13$ and $S(6) = 9/5$, leaving only $b = 1$ in each case. The discriminants are 221 and 313, respectively. Since $14^2 < 221 < 15^2$ and $17^2 < 313 < 18^2$, both give irrational antisymmetric eigenvalues. Every larger b is covered by Theorem 6.5. $\square$

Corollary 11.3. *Every positive b also gives a nonintegral spectrum for $a = 7, 8, 9$. Thus every repeated exponent $a \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ is settled.*

Proof. The third cutoff gives $T(7) = 6/5$, $T(8) = 12/7$, $T(9) = 70/31$. Only four pairs remain under these bounds:

$$(a, b) = (7, 1), (8, 1), (9, 1), (9, 2).$$

Their discriminants are 421, 545, 685, 1489, respectively, between the consecutive squares with lower square roots 20, 23, 26, 38. Each therefore gives irrational antisymmetric eigenvalues, and Theorem 6.7 covers every larger b . $\square$

For comparison, the complete divisor certificates before applying the linear cutoff are given in Table 2. The constants factor as $M = 40 = 2^3 \cdot 5$, $189 = 3^3 \cdot 7$, and $576 = 2^6 \cdot 3^2$. Testing every divisor $d \leq \sqrt{M}$ in (5) gives exactly the displayed admissible pairs. The corresponding cubics in Table 3 have no root modulo the listed primes. These finite certificates supply a second proof using the complete exceptional list. Three pairs are in the boundary family; $(4, 2)$ is the additional symmetric-block obstruction.

a	M	(d, e)	b	$\sqrt{D}$
2	40	(1, 40)	3	13
3	189	(1, 189)	10	47
4	576	(6, 96)	2	18
4	576	(1, 576)	21	115

Table 2: Every admissible factor pair for $a = 2, 3, 4$.

(a, b)	$f_{a,b}(x)$	Witness prime
(2, 3)	$x^3 - 47x^2 + 702x - 3312$	5
(3, 10)	$x^3 - 187x^2 + 11220x - 214200$	13
(4, 2)	$x^3 - 86x^2 + 2304x - 18816$	5
(4, 21)	$x^3 - 485x^2 + 75492x - 3721536$	5

Table 3: Root-free modular reductions for the exceptional cubics.

The separate diagnostic range $a = 2, 3, 4$, $1 \leq b \leq 30$ contains 90 cubics, each with a noninteger root certified in an interval between consecutive integers. The reducible cases in that range are $b = 2, 6$ for $a = 2$, $b = 3, 12$ for $a = 3$, and $b = 4, 20$ for $a = 4$. Each has a linear and an irreducible quadratic factor. These observations do not require whole-cubic irreducibility. Indeed, for every $a \geq 1$,

$$\begin{aligned} f_{a,a}(x) &= (x - 2a(a+1))(x^2 - (5a^2 + 2a)x + 6a^4 + 3a^3), \\ f_{a,a(a+1)}(x) &= (x - 2a(a+1)^2)(x^2 - (3a^3 + 4a^2 + 2a)x + 2a^6 + 6a^5 + 7a^4 + 3a^3). \end{aligned}$$

Consequently no prime can make every diagnostic cubic irreducible: the monic factors in the diagonal family retain positive degree modulo every prime.

For the second family the quadratic discriminant is $a^2(a^4 + 4a + 4)$. For $a \geq 3$ the radicand is between a^4 and $(a^2 + 1)^2$, since $2a^2 - 4a - 3 = 2(a - 3)^2 + 8(a - 3) + 3 > 0$. At $a = 2$ it is 28, between 5^2 and 6^2 . Thus the quadratic is irreducible for every $a \geq 2$.

Remark 11.4. At $a = 1, b = 2$ the cubic is $(x - 3)(x - 6)(x - 8)$, although the full graph has irrational antisymmetric eigenvalues $(13 \pm \sqrt{33})/2$. A direct construction of its 10 vertices gives

$$\det(xI - L) = x(x - 10)(x - 9)^3(x - 7)^2(x - 4)(x^2 - 13x + 34).$$

Thus a noninteger cubic root is a sufficient condition for graph nonintegrality; the converse cannot be used without the other blocks.

12 Exact verification and reproducibility

The infinite-family conclusions follow from the proofs above, including the explicitly exhausted finite certificates in the repeated-exponent and minimum-three arguments. The accompanying computations use integer or rational arithmetic to verify embeddings, identities and finite certificates. In particular, the all- b result for $a = 2, 3, 4$ uses the exhaustive divisor criterion, rather than extrapolation from the 90 diagnostic cases. No Lean formalization is claimed for these Laplacian results.

The code and saved JSON certificates are available in the public project repository. The build instructions beside this manuscript give commands and record the exact source revisions. The checks have the following scopes.

1. `scripts/check_integrality_obstructions.py` constructs ideals as exponent tuples and tests adjacency by coordinatewise maxima, independently of the support description. It verifies $LU = UB$ for the displayed invariant embeddings, for squarefree $t = 3, \dots, 8$ and $(1, 1, k)$ with $k = 1, \dots, 8$: 14 cases and 44 294 vertex pairs. It also checks the square-gap inequalities and principal-minor witnesses.
2. `scripts/check_remaining_case.py` constructs all 34 vertices for $(2, 2, 3)$ and verifies its full characteristic polynomial and the class decomposition in exact arithmetic. The shifted symmetric cubic is $x^3 - 80x^2 + 2099x - 18052$; its reduction modulo 5 has no root.
3. `scripts/check_repeated_exponent_family.py` checks the general quotient polynomial, both boundary sign identities, the positive-coefficient expansion, and the integral antisymmetric eigenvalues symbolically. Direct graphs for $(2, 2, 3)$, $(3, 3, 10)$, $(4, 4, 21)$ have 34, 174, 548 vertices, covering 165 490 vertex pairs. Exact neighbor counts verify both blocks and their lifting. The check computes the seven-support quotient polynomial; it does not compute the full 548×548 characteristic polynomial.

4. `scripts/check_aa_b.py` factors the 90 diagnostic cubics over $\mathbb{Q}$ and uses exact rational root isolation for a root between consecutive integers. Independent Sturm counts and endpoint signs were checked for every saved interval. Its modular search is restricted to 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31; absence of a listed witness is not a proof that no prime works for that cubic.
5. `scripts/check_fixed_repeated_exponents.py` reconstructs the cubic from B , checks the discriminant, product, cutoff and reducible factor identities, and enumerates every factor pair for $a = 2, 3, 4$. It verifies the admissibility conditions and all modular witnesses in Tables 2 and 3. The general congruence and factor inequality are proved in Section 6.
6. `scripts/check_second_cutoff.py` checks the polynomial remainder modulo $a + 2$, enumerates every divisor of 24, and verifies the four root-free cubic reductions settling all $d = a + 2$ cases. It also checks the second-cutoff identities and the two nonsquare discriminants completing $a = 5, 6$. This finite certificate follows the proved divisor reduction; it is not a parameter-range search.
7. `scripts/check_factor_index_reduction.py` verifies the general fixed-index divisibility and reconstruction identities, exhausts the divisors of $K_2 = 108$, and certifies the unique positive- b index-2 cubic modulo 13. It checks the third-cutoff identities and the four remaining discriminants for $a = 7, 8, 9$. These finite checks exhaust the proved candidate sets, with no arbitrary parameter cutoff.
8. `scripts/check_repeated_exponent_completion.py` reconstructs the general cubic from the 4×4 quotient, checks both sign-interval identities, and verifies the fourth-cutoff and strictly positive gap identities. It exhausts every divisor of 24, 108, 320 for indices 1, 2, 3, reproducing all eight positive- b modular cubic certificates with separate integer Horner arithmetic. The infinite remaining range is settled by the sign proof of Theorem 7.1, not by a scan.
9. `scripts/check_six_support_quotient.py` verifies the general six-support and complement quotients, their weighted symmetry and reflected quintics, and the two determinant values used in Theorem 8.1. Exact rational factorization covers all thirty-five requested triples $1 \leq a < b < c \leq 7$, with every matrix and quintic cross-checked against `scripts/check_fully_distinct.py`. All saved rational intervals pass independent Sturm counts and endpoint-sign checks. Direct ideal adjacency verifies the quotient for $(1, 2, 3)$ and $(2, 3, 4)$, using 22 and 58 vertices and 312 representative-to-vertex pairs. The infinite unit-exponent conclusion uses the written endpoint proof.
10. `scripts/check_minimum_two.py` verifies the complement endpoint identities at $a = 2$, the symmetric expression in $b + c$ and bc , and both positive-coefficient expansions that exhaust the infinite range. It reconstructs the single exceptional quintic at $(2, 3, 4)$ and checks its rational endpoint values with separate Horner evaluation and a Sturm count. No new parameter range is enumerated.
11. `scripts/check_distinct_tail.py` reconstructs the general six-support complement quintic and checks its cubic first endpoint, the uniform cutoff identity and positive remainder, and both minimum-three second-endpoint expansions. The cutoff itself derives the finite domain $4 \leq b < c < 30$, containing exactly 325 pairs. For every pair the checker reconstructs the integer quotient quintic, matches its symbolic specialization, and checks the displayed root interval using separate rational Horner arithmetic. Separate cubic Horner evaluation confirms

all first-endpoint values: 257 positive, 68 negative and no zero. The $(3, 4, 5)$ rational exception also has a Sturm count of one in $(1, 3/2)$. The minimum-three theorem uses this complete finite certificate; the general cutoff uses the written positive-coefficient argument.

The symbolic and characteristic-polynomial checks use SymPy 1.14.0; the first checker uses the Python standard library. Saved results contain source hashes. The 10-vertex polynomial in the preceding remark was independently checked by a direct ideal-graph construction; it is also reproducible by the command in the manuscript build guide. No floating-point eigenvalue calculation is used as integrality evidence.

13 Remaining questions

Theorem 7.1 settles every triple with at least two equal exponents, and Theorem 10.2 settles every triple with minimum exponent at most three. Proposition 10.1 settles every ordered triple with $c \geq 4a^2 - 2a$. The remaining three-prime cases therefore have $4 \leq a < b < c < 4a^2 - 2a$. These cases and nonsquarefree exponent vectors with more than three prime factors require further arguments. The results here therefore do not establish nonintegrality for every integer with at least three distinct prime factors.

The remaining three-prime question is whether $G_{p^a q^b r^c}$ is always nonintegral in this remaining region. For each fixed minimum exponent a , the cutoff leaves a finite set of pairs; it does not bound a itself. The coordinate-swap decomposition used for repeated exponents is unavailable there; the full six-support quotient provides an exact starting point.

The distinction between the two repeated-exponent blocks remains essential: an integral anti-symmetric block occurs on the boundary family, while an integral symmetric cubic occurs at $(1, 1, 2)$. The completion theorem uses both blocks, rather than asserting that either alone always obstructs integrality.

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